

nant wireless charging and offers an insight into a world where wireless power is widely integrated into the home, office and everyday lives:

1. From an environmental standpoint, wireless charging could be incredibly promising. It would eliminate the need for plastic-body chargers, which not only drain power but are also replaced regularly. Wireless charging could save tens of thousands of tons of chargers from ending up in landfills and allow users to tap into existing electronic activity instead of traditional power sources.

2. It provides greater convenience and ubiquity for charging of everyday devices.

3. It would reduce costs associated with maintaining mechanical connectors.

4. It would allow safe powering or charging of devices that need to remain sterile or hermetically-sealed (waterproof).

5. It would prevent corrosion due to elements such as oxygen and water.

6. It would eliminate sparks and debris associated with wired contacts.

7. It would provide protected connections; no corrosion when electronics are all enclosed, away from water or oxygen in the atmosphere.

8. It is safer for medical implants (embedded medical devices). It allows recharging/powering through skin rather than having wires penetrate the skin, which increases the risk of infection.

9. It is durable. Without the need to constantly plug and unplug devices, there is significantly less wear and tear on the sockets of the devices and the attaching cables.

10. It enables non-radiative energy transfer.

11. It can be more convenient as there is no need for cables.

Disadvantages

Some disadvantages of wireless charging are:

Lower efficiency and waste heat. Main disadvantages of inductive charging are its lower efficiency and increased resistive heating in comparison to direct contact. Implementations using lower frequencies or older drive technologies charge more slowly and generate heat within most portable electronics. It is not as efficient as a direct cable connection. Wireless charging is around 60 per cent to 70 per cent efficient, and it is still recommended that booting a device from cold is done through a wired connection.

Slower charging. Due to lower efficiency, devices can take longer to charge when supplied similar power.

More expensive. Inductive charging also requires drive electronics and coils in both device and charger, increasing complexity and cost of manufacturing.

Short-distance applications. Whether or not it incorporates resonance, induction generally sends power over relatively short distances.

Solutions

Given below are some possible solutions:

1. Newer approaches reduce transfer losses through the use of ultra-thin coils, higher frequencies and optimised drive electronics. This results in more efficient and compact chargers and receivers, facilitating their integration into mobile devices or batteries with minimal changes required. These technologies provide charging times comparable to wired approaches, and these are rapidly finding their way into mobile devices.

2. For day-to-day use, just lining up the coils and letting electromagnetism do the rest is the simple value proposition at the heart of wireless charging. It is worth noting that the lithium-ion chemistry used in smartphone batteries happily copes with having short bursts of energy to charge these up. This is exactly what wireless charging can provide.

3. Future plans for wireless power involve moving electricity over a span of kilometres. A few proposals even involve sending power to the Earth from space. The secret was a large, ground-based microwave transmitter. A large, disk-shaped rectifying antenna, or rectenna, changed the microwave energy from the transmitter into DC electricity. This was because the microwaves' interaction with the rectenna had a constant power supply as long as it was in the range of a functioning microwave array.

Rectifying antennae are central to many wireless power-transmission theories. These are usually made of an array of dipole antennae, which have positive and negative poles. These antennae connect to semiconductor diodes. These collect microwave energy and transmit it to the diodes, which act like switches that are open or closed as well as turnstiles that let electrons flow in only one direction. These direct the electrons to the rectenna's circuitry. The circuitry routes the electrons to the parts and systems that need these.

Safety concerns

There will be safety issues, real or imagined, as all power has to pass through space in some form or the other, and pass through any bodies lying in its path. However, researchers have minimised this problem by ensuring that power is mainly in the form of a magnetic field, a form of energy to which the body is almost entirely insensitive. It is believed that wireless charging is safe, even for people with implanted medical devices such as pacemakers.

Although researchers have not made a detailed study to test how the system interferes with pacemakers, however, they do not expect it to interact strongly with objects that do not resonate at the same frequencies that are used to transfer power. **EFY**